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VU Answer

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Al-Khwarizmi's work was written in a book titled _____

Answer (Please select your correct option)



al Kitab al-mukhtasar fi hisab al-jabrwa'l-muqabalah



Calculation by Completion



al Kitab



al-jabrwa'l-muqabalah



What is the asymptotic growth of $\frac{4n^3 + 15n^2 + 11n}{6}$?

Answer (Please select your correct option)

☐ $\Theta\left(\frac{4n^3 + 15n^2 + 11n}{6}\right)$

☐ $\Theta(4n^3 + 15n^2 + 11n)$

☒ $\Theta(15n^2)$

☐ $\Theta(n^3)$





The reason for introducing Sieve Technique algorithm is that it illustrates a very important special case of,

Answer (Please select your correct option)



divide-and-conquer



decrease and conquer



greedy nature



2-dimension Maxima



Strictly increasing arrays are _____ input for quick sort

Answer (Please select your correct option)

☐

The worst-case

☐

The best-case

☐

Average case

☐

None of the given



The running time of quick sort depends heavily on the selection of

Answer (Please select your correct option)

☐ Required Output

☐ Input and output

☒ Pivot

☐ Input data only





Quick sort is based on divide and conquer paradigm. In the Combine step of divide and conquer process

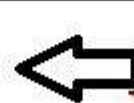
Answer (Please select your correct option)

☐ No work is needed to combine the sub-arrays, the array is already sorted

☐ Work is needed to combine the sub-arrays; the array is not already sorted.

☐ Merging the Arrays

☐ Dividing the arrays



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In in-place sorting algorithm is one that uses no _____ arrays for storage.

Answer (Please select your correct option)



two dimensional



three dimensional



n dimensional



additional



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Bubble sort and insertion sort can be implemented as stable algorithms but _____ can not be.

Answer (Please select your correct option)



selection sort

correct answer solve
by hadi



bubble sort



merge sort



stable sort



Catalan number is given by the formula

Answer (Please select your correct option)

☐ $C(n) = \frac{1}{n+1} \binom{2n}{n}$

correct answer solve
by hadi

☐ $C(n) = \frac{1}{n-1} \binom{2n}{n}$

☐ $C(n) = \frac{1}{n+1} \binom{n}{2n}$

☐ $C(n) = \frac{1}{n-1} \binom{n}{2n}$



We can find the product $A \times B$ of matrices A and B, only if they are compatible which means,

Answer (Please select your correct option)

☐

No of Columns of A must be equal to No of Rows of B

☐

No of Columns of A must be equal to No of Columns of B

☐

No of Rows of A must be equal to No of Rows of B

correct answer solve
by hadi

☐

Order of A must be equal to order of B



Matrix – Chain – Order is _____ than the exponential time method of enumerating all possible parenthesizations and checking each one.

Answer (Please select your correct option)



Much more efficient



Only fractional efficient



Worst



Too slow



Time complexity of chain matrix multiplication is $\Theta(n^3)$ and space complexity is

Answer (Please select your correct option)



$\Theta(n^2)$

correct answer solve
by hadi



$\Theta(n^3)$



$\Theta(n \log n)$



$\Theta(\log n)$



The Knapsack problem belongs to the domain of _____ problems.

Answer (Please select your correct option)



Optimization

correct answer solve
by hadi



NP Complete



Linear Solution



Sorting



The recurrence relation of Tower of Hanoi is given below

$$T(n) = \begin{cases} 1 & \text{if } n=1 \\ 2T(n-1) + 1 & \text{if } n > 1 \end{cases}$$

In order to move a tower of 4 rings from one peg to another, how many ring moves are required?

Answer (Please select your correct option)

☐

15

☐

7

☐

12

☐

32



Which type of instructions Random Access Machine (RAM) can execute? Choose best answer

Answer (Please select your correct option)



Algebraic and logic



Geometric and arithmetic



Arithmetic and logic

correct answer solve
by hadi



Parallel and recursive



Algorithm's essential elements are

Answer (Please select your correct option)

☐

Step wise solution

☐

Stepwise solution and finite time

☐

Step wise solution finite inputs

correct answer solve
by hadi

☐

Stepwise approach in which time and memory does not matter.

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Which one is the correct conversion of the given algorithm to sigma notation?

MAXIMA(int n, Point P[1...n])

1 for i ← 1 to n n times

2 do maximal ← true

3 end for

Answer (Please select your correct option)

☐ $\sum_{i=1}^n 4n + 4$

☐ $\sum_{i=1}^n 4n + 2$

correct answer solve
by hadi

☐ $\sum_{i=1}^n 4n + 2n$

☐ $\sum_{i=1}^n 2n + 2$



Which of the following is calculated with **Big Theta** notation?

Answer (Please select your correct option)



Lower bounds



Upper bounds

correct answer solve
by hadi



Both upper and lower bound



Medium bounds



Suppose we have an algorithm that carries out N^2 operations for an input of size N . Let us say that a computer takes 1 microsecond ($1/1000000$ second) to carry out one operation. How long does the algorithm run for an input of size 3000?

Answer (Please select your correct option)



90 seconds



9 seconds

correct answer solve
by hadi



0.9 seconds



0.09 seconds

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Which one is the best algorithm from the following with respect to running time?

Answer (Please select your correct option)



$\Theta(n)$



$\Theta(n^2)$

correct answer solve
by hadi



$\Theta(n \log n)$



$\Theta(n^3)$



Unlike programs, algorithms to be understood primarily by _____ and _____

Answer (Please select your correct option)



Machines, not people



Mathematical expressions, not algebraic expressions



Programmers, not machines

correct answer solve
by hadi



RAM, not programmer

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Recurrences are useful for analyzing

Answer (Please select your correct option)



Recursive Algorithms

correct answer solve
by hadi



Simple Algorithms



Parallel Algorithms



Parallel Algorithms & Recursive Algorithms

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Divide-and-conquer involves breaking the problem into a small number of

Answer (Please select your correct option)

☐ pivot

☐ Sub problems

correct answer solve
by hadi

☐ Selection

☐ Sieve

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In which order we can sort?

Answer (Please select your correct option)

☐ increasing order or decreasing order

correct answer solve
by hadi

☐ both at the same time

☐ increasing order only

☐ decreasing order only

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Comparison based sorting algorithms can not run faster than

Answer (Please select your correct option)



$\Omega(n \log n)$

correct answer solve
by hadi



$O(n \log n)$



$\Omega(n^2)$



$O(n^2)$

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In Quick sort, we don't have the control over the sizes of recursive calls

Answer (Please select your correct option)



True

correct answer solve
by hadi



False



Less information to decide



Either true or false

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Who invented Quick sort procedure?

Answer (Please select your correct option)



Hoare

correct answer solve
by hadi



Sedgewick



Mellroy



Coreman

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Counting sort assumes that the numbers to be sorted are in the range 1 to k , where k is

Answer (Please select your correct option)



Small

correct answer solve
by hadi



Large



No restriction on k



None of these



If there are $\Theta(n^2)$ entries in edit distance matrix then the total running time is

Answer (Please select your correct option)

☐ $\Theta(1)$ ☐ $\Theta(n^2)$

correct answer solve
by hadi

☐ $\Theta(n)$ ☐ $\Theta(n \log n)$



When a recursive algorithm revisits the same problem over and over again, we say that the optimization problem has _____ sub-problems.

Answer (Please select your correct option)

☐ Overlapping

☐ Over costing

☐ Optimized

☐ None of these



A $p \times q$ matrix A can be multiplied with a $q \times r$ matrix B . The result will be a $p \times r$ matrix C . In particular, for $1 \leq i \leq p$ and $1 \leq j \leq r$,

Answer (Please select your correct option)

☐ $C[i, j] = \sum_{k=1}^q A[i, k]B[k, j]$



☐ $C[i, j] = \sum_{k=1}^q A[k, i]B[k, j]$

☐ $C[i, j] = \sum_{k=1}^q A[k, i]B[j, k]$

☐ None of these



We can find the product $A \times B$ of matrices A and B, only if they are compatible which means,

Answer (Please select your correct option)

☐

No of Columns of A must be equal to No of Rows of B

☐

No of Columns of A must be equal to No of Columns of B

☐

No of Rows of A must be equal to No of Rows of B

correct answer solve
by hadi

☐

Order of A must be equal to order of B

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Time complexity of chain matrix multiplication is $\Theta(n^3)$ and space complexity is

Answer (Please select your correct option)

☐

$\Theta(n^2)$

correct answer solve
by hadi

☐

$\Theta(n^3)$

☐

$\Theta(n \log n)$

☐

$\Theta(\log n)$



Computational model of sequential RAM is:

Answer (Please select your correct option)

☐

Parallel machines may be expensive to model and have more computational power than sequential RAM.

☐

Computational power of sequential RAM is same as that of parallel machines only time efficiency is achieved with parallel machines

☐

Both first and second options are true for the statement

☐

Less powerful computational wise than parallel machines

correct answer solve
by hadi

Made By: Waqar Siddhu

The worst case running time of the algorithm given below is,

```
MAXIMA(int n, Point P[1...n])
```

```
1  for i ← 1 to n n times
```

```
2  do maximal ← true
```

```
3  ...
```

Answer (Please select your correct option)

☐ $\Theta(n^6)$

☐ $\Theta\left(\frac{2n}{n^6}\right)$

☐ $\Theta(n^2)$

correct answer solve
by hadi

☐ $\Theta(2n \lg 6)$

```
1  for i ← 1 to n      n times
2  do maximal ← true
3    for j ← 1 to n    n times
4    do
5      if (i ≠ j) & (P[i].x < P[j].x) & (P[i].y < P[j].y)  4 accesses
```

Answer (Please select your correct option)

☐

$$\Theta(n^6)$$

☐

$$\Theta\left(\frac{2n}{n^6}\right)$$

☐

$$\Theta(n^2)$$

correct answer solve
by hadi

☐

$$\Theta(2n \lg 6)$$

```
3   for j ← 1 to n n times
4   do
5       if (i ≠ j) & (P[i].x ≤ P[j].x) & (P[i].y ≤ P[j].y) 4 accesses
6           then maximal ← false break
7   if maximal _____
```

Answer (Please select your correct option)

☐

$$\Theta(n^6)$$

☐

$$\Theta\left(n^{\frac{2n}{6}}\right)$$

☐

$$\Theta(n^2)$$

correct answer solve
by hadi

☐

$$\Theta(2n \lg 6)$$

```
5      if (i ≠ j) & (P[i].x ≤ P[j].x) & (P[i].y ≤ P[j].y) 4 accesses
6          then maximal ← false break
7      if maximal
8          then output P[i].x, P[i].y 2 accesses
```

Answer (Please select your correct option)

☐

$$\Theta(n^6)$$

☐

$$\Theta\left(n^{\frac{2n}{6}}\right)$$

☐

$$\Theta(n^2)$$

correct answer solve
by hadi

☐

$$\Theta(2n \lg 6)$$



In Sieve Technique, we know _____

Answer (Please select your correct option)



Item of interest



Order of items



complexity of items



All items are of interest

correct answer solve
by hadi

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In Random access machine, instructions are executed _____.

Answer (Please select your correct option)

☐ five at a time

☐ infinite instructions at a time

☐ one-by-one

☐ parallel

correct answer solve
by hadi

Made By: Waqar Siddhu



Which type of instructions Random Access Machine (RAM) can execute? Choose best answer

Answer (Please select your correct option)



Algebraic and logic



Geometric and arithmetic



Arithmetic and logic

correct answer solve
by hadi



Parallel and recursive

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Which of the following is calculated with **Big O notation**?

Answer (Please select your correct option)



Medium bounds



Lower bounds



Upper bounds

correct answer solve
by hadi



Both upper and lower bound

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Which of the following functions grows fastest as n gets larger?

Answer (Please select your correct option)



$$n^{10}2^n$$



$$n^83^n$$



$$n^55^n$$



$$n^32^{2n}$$



Al-Khwarizmi's work was written in a book titled _____

Answer (Please select your correct option)

☐ al Kitab al-mukhtasar fi hisab al-jabrwa'l-muqabalah

correct answer solve
by hadi

☐ Calculation by Completion

☐ al Kitab

☐ al-jabrwa'l-muqabalah



Unlike programs, algorithms to be understood primarily by _____ and _____

Answer (Please select your correct option)

☐

Machines, not people

☐

Mathematical expressions, not algebraic expressions

☐

Programmers, not machines

correct answer solve
by hadi

☐

RAM, not programmer

Made By: Waqar Siddhu

Which formula is used for calculating worst case running time?

Answer (Please select your correct option)

☐

$$T_{worst}(n) = \max_{|I|=n} T(I)$$

correct answer solve
by hadi

☐

$$T_{worst}(n) = \max_{|I|=1} T(I)$$

☐

$$T_{worst}(n) = \max_{|I|=1} T(n)$$

☐

$$T_{worst}(n) = \max_{|I|=n} T(n)$$

Analysis of Selection algorithm ends up with,

page 37

Answer (Please select your correct option)

☐ $\Theta(n)$ correct answer solve
by hadi

☐ $\Theta(1 / 1 + n)$

☐ $\Theta(n / 2)$

☐ $\Theta((n / 2) + n)$

A (an) _____ is a left-complete binary tree that conforms to the *heap order*

p 40

Answer (Please select your correct option)

☐ heap

correct answer solve
by hadi

☐ binary tree

☐ binary search tree

☐ array

Principal operation for maintaining the heap property is called heapify, it is also called

p 43

Answer (Please select your correct option)

☐ sifting down

correct answer solve
by hadi

☐ sifting up

☐ sifting left

☐ sifting right

Made By: Waqar Siddhu



What is common between Bubble sort, Insertion sort, Selection sort, Quick sort, and Heap sort?

Answer (Please select your correct option)

☐

All are in-place algorithms

correct answer solve
by hadi

☐

All are stable algorithms

☐

None of these

☐

All are unstable algorithms

Made By: Waqar Siddhu

When primary memory storage is at a premium, a (an) _____ algorithm may be preferable.

:
:(

Answer (Please select your correct option)

- ☐ In-place
- ☐ Stable
- ☐ External storage
- ☐ None of these

correct answer solve
by hadi

Made By: Waqar Siddhu

If there are $\Theta(n^2)$ entries in edit distance matrix then the total running time is

Answer (Please select your correct option)

☐ $\Theta(1)$

☐ $\Theta(n^2)$

correct answer solve
by hadi

☐ $\Theta(n)$

☐ $\Theta(n \log n)$

Catalan number is given by the formula

Answer (Please select your correct option)

☐ $C(n) = \frac{1}{n+1} \binom{2n}{n}$

correct answer solve
by hadi

☐ $C(n) = \frac{1}{n-1} \binom{2n}{n}$

☐ $C(n) = \frac{1}{n+1} \binom{n}{2n}$

☐ $C(n) = \frac{1}{n-1} \binom{n}{2n}$



For Chain Matrix Multiplication we can not use divide and conquer approach because,

Answer (Please select your correct option)



We do not know the optimum k

correct answer solve
by hadi



We use divide and conquer for sorting only



We can easily perform it in linear time



Size of data is not given

We can find the product $A \times B$ of matrices A and B, only if they are compatible which means,

Answer (Please select your correct option)

☐

No of Columns of A must be equal to No of Rows of B

☐

No of Columns of A must be equal to No of Columns of B

☐

No of Rows of A must be equal to No of Rows of B

correct answer solve
by hadi

☐

Order of A must be equal to order of B

Made By: Waqar Siddhu



Time complexity of chain matrix multiplication is $\Theta(n^3)$ and space complexity is

Answer (Please select your correct option)

☐ $\Theta(n^2)$ correct answer solve by hadi

☐ $\Theta(n^3)$

☐ $\Theta(n \log n)$

☐ $\Theta(\log n)$

The worst case running time of the algorithm given below is,

```
MAXIMA(int n, Point P[1...n])
```

```
1  for i ← 1 to n n times
```

```
2  do maximal ← true
```

Answer (Please select your correct option)

☐ $\Theta(n^6)$

☐ $\Theta\left(\frac{2n}{n^6}\right)$

☐ $\Theta(n^2)$

correct answer solve
by hadi

☐ $\Theta(2n \lg 6)$

MAXIMA(int n, Point P[1...n])

1 for i ← 1 to n n times

2 do maximal ← true

3 for j ← 1 to n n times

4 do

Answer (Please select your correct option)

☐ $\Theta(n^6)$

☐ $\Theta\left(n^{\frac{2n}{6}}\right)$

☐ $\Theta(n^2)$

☐ $\Theta(2n \lg 6)$

already submit,,,,,correct
answer solve by hadi

Made By: Waqar Siddhu

```
3   for j ← 1 to n n times
4   do
5       if (i ≠ j) & (P[i].x ≤ P[j].x) & (P[i].y ≤ P[j].y) 4 accesses
6           then maximal ← false break
7   if maximal
```

Answer (Please select your correct option)

☐ $\Theta(n^6)$

☐ $\Theta\left(n^{\frac{2n}{6}}\right)$

☐ $\Theta(n^2)$

☐ $\Theta(2n \lg 6)$

correct answer solve
by hadi...already
submit

```
5      if (i ≠ j) & (P[i].x ≤ P[j].x) & (P[i].y ≤ P[j].y) 4 accesses
6          then maximal ← false break
7      if maximal
8          then output P[i].x, P[i].y 2 accesses
```

Answer (Please select your correct option)

☐ $\Theta(n^6)$

☐ $\Theta\left(n^{\frac{2n}{6}}\right)$

☐ $\Theta(n^2)$

correct answer solve
by hadi....already
submit

☐ $\Theta(2n \lg 6)$

In Sieve Technique, we know _____

it applies to problems where we are interested in finding

a single item from a larger set of n items. We do not know which item is of interest,

repeat

Answer (Please select your correct option)

☐ Item of interest

☐ Order of items

☐ complexity of items

☐ All items are of interest

correct answer solve
by hadi

Made By: Waqar Siddhu



In Random access machine, instructions are executed _____.

Answer (Please select your correct option)



five at a time



infinite instructions at a time



one-by-one

correct answer solve
by hadi



parallel

Made By: Waqar Siddhu



Which of the following is calculated with **Big Theta** notation?

Answer (Please select your correct option)

☐

Lower bounds

☐

Upper bounds

correct answer solve
by hadi

☐

Both upper and lower bound

☐

Medium bounds

Made By: Waqar Siddhu



Suppose we have an algorithm that carries out N^2 operations for an input of size N . Let us say that a computer takes 1 microsecond ($1/1000000$ second) to carry out one operation. How long does the algorithm run for an input of size 3000?

Answer (Please select your correct option)



90 seconds



9 seconds

correct answer solve
by hadi



0.9 seconds



0.09 seconds

Which one is the best algorithm from the following with respect to running time?

Answer (Please select your correct option)

- ☐ $\Theta(n)$
- ☐ $\Theta(n^2)$ correct answer solve by hadi
- ☐ $\Theta(n \log n)$
- ☐ $\Theta(n^3)$



Which of the following is true?

Answer (Please select your correct option)

☐ The worst case time complexity of the quick sort is $N \log N$

☐ The best case time complexity of the merge sort is $\log N$

☐ The worst case time complexity of the selection sort is N^2

☐ The worst case time complexity of the merge sort is $\log N$

correct answer solve
by hadi

Made By: Waqar Siddhu

Question No : 1 of 26

Marks: 1 (Budgeted Time 1 Min)



The word *Algorithm* comes from the name of the Muslim author _____



Answer (Please select your correct option)

- ☐ Abu Ja'far Mohammad ibn Musaal-Khowarizmi or Al-Khwarizmi correct answer solve by hadi
- ☐ Khwarizm
- ☐ Kheva
- ☐ Uzbekistan

Start Time: 5:37 AM
60:00
Time Left



1



Question No : 2 of 26

Marks: 1 (Budgeted Time 1 Min) 

Al-Khwarizmi's work was written in a book titled _____

Answer (Please select your correct option)

☐ al Kitab al-mukhtasar fi hisab al-jabrwa'l-muqabalah

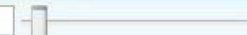
correct answer solve
by hadi

☐ Calculation by Completion☐ al Kitab☐ al-jabrwa'l-muqabalah

Start Time: 5:37 AM

60:00

Time Left

2 

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File Edit View Favorites Tools Help

Favorites Suggested Sites Web Slice Gallery

VU Examination System (CLIENT) VUTES 6.5 :..... Fall...

Live Search

Home RSS Mail Print Page Safety Tools

Question No : 3 of 26

Marks: 1 (Budgeted Time 1 Min)

_____ is a graphical representation of an algorithm

Answer (Please select your correct option)

- ☐ Σ notation
- ☐ Θ notation
- ☐ Flowchart
- ☐ Asymptotic notation

correct answer solve
by hadi

Start Time: 5:37 AM

60:00

Time Left



3



Question No : 4 of 26

Marks: 1 (Budgeted Time 1 Min) 

What is the asymptotic growth of $n^2 + \frac{4n + 15n + 11n}{16}$?

Answer (Please select your correct option)

☐ $\Theta\left(\frac{4n + 15n + 11n}{16}\right)$

☐ $\Theta\left(n^2 + n + \frac{1}{16}\right)$

☐ $\Theta(n^2)$

correct answer solve
by hadi

☐ $\Theta(n)$

Start Time: 5:37 AM

60:00

Time Left



4

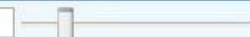


Question No : 5 of 26

Marks: 1 (Budgeted Time 1 Min) 

Recurrences are useful for analyzing

Answer (Please select your correct option)

☐ Recursive Algorithmscorrect answer solve
by hadi☐ Simple Algorithms☐ Parallel Algorithms☐ Parallel Algorithms & Recursive AlgorithmsStart Time: 5:37 AM
60:00
Time Left 5 

Question No : 6 of 26

Marks: 1 (Budgeted Time 1 Min) 

The sieve technique is a special case, where the number of sub problems is just

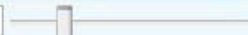
Answer (Please select your correct option)

- ☐ 5
- ☐ many
- ☐ 1
- ☐ few
- correct answer solve
by hadi

Start Time: 5:37 AM

60:00

Time Left

6 

Question No : 7 of 26

Marks: 1 (Budgeted Time 1 Min) 

What is the total time to heapify?

Answer (Please select your correct option)

☐ $O(\log n)$ correct answer solve by hadi

☐ $O(n \log n)$

☐ $O(n^2 \log n)$

☐ $O(\log^2 n)$

Start Time: 5:37 AM

60:00

Time Left



7



Question No : 8 of 26

Marks: 1 (Budgeted Time 1 Min) 

Comparison based sorting algorithms can not run faster than

Answer (Please select your correct option)

☐ $\Omega (n \log n)$

correct answer solve
by hadi

☐ $O (n \log n)$ ☐ $\Omega (n^2)$ ☐ $O (n^2)$

Start Time: 5:37 AM

60:00

Time Left

8 

Question No : 9 of 26

Marks: 1 (Budgeted Time 1 Min)

After partitioning array in Quick sort, pivot is placed in a position such that

Answer (Please select your correct option)

☐ Values smaller than pivot are on left and larger than pivot are on right☐ Values larger than pivot are on left and smaller than pivot are on right

correct answer solve
by hadi

☐ Pivot is the first element of array☐ Pivot is the last element of array

Start Time: 5:37 AM

60:00

Time Left



9



Question No : 10 of 26

Marks: 1 (Budgeted Time 1 Min) 

The running time of quick sort depends heavily on the selection of

Answer (Please select your correct option)

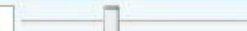
☐ Required Output☐ Input and output☐ Pivot☐ Input data only

correct answer solve
by hadi

Start Time: 5:37 AM

60:00

Time Left

10 

Question No : 11 of 26

Marks: 1 (Budgeted Time 1 Min) 

Memoization is a process

Answer (Please select your correct option)

☐ To avoid unnecessary repetitions by writing down the results of recursive calls and looking them again if needed latercorrect answer solve
by hadi☐ To avoid repeated data by writing down the results of input array and looking them again if needed later.☐ None of these☐ All options are correct

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Question No : 12 of 26

Marks: 1 (Budgeted Time 1 Min)



A problem exhibits _____ if an optimal solution to the problem contains within it optimal solution to sub-problems.

Answer (Please select your correct option)

☐ Optimal structure

correct answer solve
by hadi

☐ Efficient structure☐ Inefficient structure☐ Unknown behavior

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Question No : 13 of 26

Marks: 1 (Budgeted Time 1 Min) 

A $p \times q$ matrix A can be multiplied with a $q \times r$ matrix B. The result will be a $p \times r$ matrix C. There are $(p \cdot r)$ total entries in C and each takes _____ to compute.

Answer (Please select your correct option)

☐ (q) correct answer solve by hadi

☐ (1)

☐ (n^2)

☐ (n^3)

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Question No : 14 of 26

Marks: 1 (Budgeted Time 1 Min)

Catalan number is given by the formula

Answer (Please select your correct option)

☐
$$C(n) = \frac{1}{n+1} \binom{2n}{n}$$

correct answer solve
by hadi

☐
$$C(n) = \frac{1}{n-1} \binom{2n}{n}$$

☐
$$C(n) = \frac{1}{n+1} \binom{n}{2n}$$

☐
$$C(n) = \frac{1}{n-1} \binom{n}{2n}$$

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Question No : 15 of 26

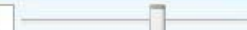
Marks: 1 (Budgeted Time 1 Min) 

For Chain Matrix Multiplication we can not use divide and conquer approach because,

Answer (Please select your correct option)

- ☐ We do not know the optimum k correct answer solve by hadi
- ☐ We use divide and conquer for sorting only
- ☐ We can easily perform it in linear time
- ☐ Size of data is not given

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Question No : 16 of 26

Marks: 1 (Budgeted Time 1 Min) 

Time complexity of chain matrix multiplication is $\Theta(n^3)$ and space complexity is

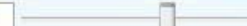
Answer (Please select your correct option)

- ☐ $\Theta(n^2)$ correct answer solve by hadi
- ☐ $\Theta(n^3)$
- ☐ $\Theta(n \log n)$
- ☐ $\Theta(\log n)$

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Question No : 17 of 26

Marks: 1 (Budgeted Time 1 Min) 

Computational model of sequential RAM is:

Answer (Please select your correct option)

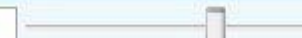
- ☐ Parallel machines may be expensive to model and have more computational power than sequential RAM.
- ☐ Computational power of sequential RAM is same as that of parallel machines only time efficiency is achieved with parallel machines
- ☐ Both first and second options are true for the statement
- ☐ Less powerful computational wise than parallel machines

correct answer solve
by hadi

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Question No : 18 of 26

Marks: 1 (Budgeted Time 1 Min) 

What is the correct order of operations in merge sort?

Answer (Please select your correct option)

☐ Conquer → Divide → Combine☐ Divide → Conquer → Combine☐ Conquer → Divide → Split☐ Split → Conquer → Destroy

correct answer solve
by hadi

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Question No : 19 of 26

Marks: 1 (Budgeted Time 1 Min) 

What is the average-case time for merge sort to sort an array of n elements?

Answer (Please select your correct option)

- ☐ $O(\log n)$
- ☐ $O(n)$
- ☐ $O(n \log n)$ correct answer solve by hadi
- ☐ $O(n^2)$

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Question No : 20 of 26

Marks: 1 (Budgeted Time 1 Min) 

What is the solution to the recurrence $T(n) = T(n/2) + n$, $T(1) = 1$?

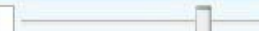
Answer (Please select your correct option)

- ☐ $O(\log n)$
- ☐ $O(n)$ correct answer solve by hadi
- ☐ $O(n \log n)$
- ☐ $O(n^2)$

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Question No : 1 of 26

Marks: 1 (Budgeted Time 1 Min) 

We analyze an algorithm to check _____

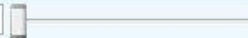
Answer (Please select your correct option)

☐ whether such an algorithm exists or not☐ its running timecorrect answer solve
by hadi☐ it is a correct algorithm or not☐ it will run on machine or not

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Question No : 2 of 26

Marks: 1 (Budgeted Time 1 Min)

What is the asymptotic growth of $\frac{8n^2}{3} + \frac{n}{5}$?

Answer (Please select your correct option)

☐ $\Theta\left(\frac{8n^2}{3}\right)$

☐ $\Theta\left(\frac{n}{5}\right)$

☐ $\Theta(n^2)$

☐ $\Theta\left(\frac{8}{3}n\right)$

correct answer solve
by hadi

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Question No : 3 of 26

Marks: 1 (Budgeted Time 1 Min)

What will be result of the following recurrence relation?

$$T(n) = \begin{cases} 4 & \text{if } n = 1 \\ T(n/5) + 3n^2 & \text{if } n \text{ is divisible by } 5 \end{cases}$$

Then T(5) is

Answer (Please select your correct option)

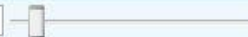
- ☐ 25
- ☐ 75
- ☐ 79
- ☐ 70

correct answer solve
by hadi

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Question No : 4 of 26

Marks: 1 (Budgeted Time 1 Min)

Divide-and-conquer involves breaking the problem into a small number of

Answer (Please select your correct option)

☐ Sieve

☐ pivot

☐ Sub problems

☐ Selection

correct answer solve
by hadi

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Question No : 5 of 26

Marks: 1 (Budgeted Time 1 Min)

The sieve technique is a special case, where the number of sub problems is just

the sieve technique is a special case, where the number of subproblems is just 1.

Answer (Please select your correct option)

☐ 5

☐ many

☒ 1

☐ few

correct answer solve
by hadi

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Question No : 6 of 26

Marks: 1 (Budgeted Time 1 Min)

When we call heapify then at each level the comparison performed takes time

Answer (Please select your correct option)

It will take $\Theta(1)$

correct answer solve by hadi

Time will vary according to the nature of input data

It can not be predicted

It will take $\Theta(\log n)$

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Question No : 7 of 26

Marks: 1 (Budgeted Time 1 Min)

In Quick sort, we don't have the control over the sizes of recursive calls

Answer (Please select your correct option)

True

False

Less information to decide

Either true or false

correct answer solve by hadi

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Question No : 8 of 26

Marks: 1 (Budgeted Time 1 Min)

The running time of quick sort depends heavily on the selection of

Answer (Please select your correct option)

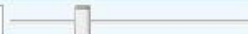
☐ Required Output☐ Input and output☐ Pivot☐ Input data only

correct answer solve
by hadi

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Question No : 9 of 26

Marks: 1 (Budgeted Time 1 Min)

Any binary tree of height $T(n)$ has at most $2^{T(n)}$ leaves. Thus a comparison based sorting algorithm can distinguish between at most _____ different final outcomes.

Answer (Please select your correct option)

☐ $2^{T(n)}$

☐ $2^{\log(n)}$

☐ 2^{n-1}

☐ 2^n

correct answer solve
by hadi

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Question No : 10 of 26

Marks: 1 (Budgeted Time 1 Min)



If there are $\Theta(n^2)$ entries in edit distance matrix then each entry $E(i, j)$ takes _____ time to compute.

Answer (Please select your correct option)

☐ $\Theta(1)$

correct answer solve
by hadi

☐ $\Theta(n^2)$ ☐ $\Theta(n)$ ☐ $\Theta(n \log n)$

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Question No : 11 of 26

Marks: 1 (Budgeted Time 1 Min) 

If there are $\Theta(n^2)$ entries in edit distance matrix then the total running time is

Answer (Please select your correct option)

☐ $\Theta(1)$ ☐ $\Theta(n^2)$ correct☐ $\Theta(n)$ ☐ $\Theta(n \log n)$

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Question No : 12 of 26

Marks: 1 (Budgeted Time 1 Min)

When a recursive algorithm revisits the same problem over and over again, we say that the optimization problem has _____ sub-problems.

Answer (Please select your correct option)

☐ Overlapping☐ Over costing☐ Optimized☐ None of these

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Question No : 13 of 26

Marks: 1 (Budgeted Time 1 Min) 

A problem exhibits optimal structure if an optimal solution to the problem contains within it optimal solution to _____

Answer (Please select your correct option)

☒ Sub-problems☐ All similar problems☐ Efficient sorting algorithm☐ Any unknown problem

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Question No : 14 of 26

Marks: 1 (Budgeted Time 1 Min) 

The Knapsack problem belongs to the domain of _____ problems.

Answer (Please select your correct option)

☐ Optimizationcorrect answer solve
by hadi☐ NP Complete☐ Linear Solution☐ Sorting

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Question No : 15 of 26

Marks: 1 (Budgeted Time 1 Min) 

Computational model of sequential RAM is:

Answer (Please select your correct option)

- ☐ Parallel machines may be expensive to model and have more computational power than sequential RAM.
- ☐ Computational power of sequential RAM is same as that of parallel machines only time efficiency is achieved with parallel machines
- ☐ Both first and second options are true for the statement
- ☐ Less powerful computational wise than parallel machines

correct answer solve
by hadi

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Question No : 16 of 26

Marks: 1 (Budgeted Time 1 Min)

Merge sort requires

Answer (Please select your correct option)

☐ Extra time than other sorting algorithms

☐ Extra storage

☐ No need of extra storage

☐ Sometimes requires extra storage

correct answer solve
by hadi

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Question No : 17 of 26

Marks: 1 (Budgeted Time 1 Min)

The worst case running time of the algorithm given below is,

```
MAXIMA(int n, Point P[1...n])
1  for i ← 1 to n
2  do maximal ← true
```

Answer (Please select your correct option)

☐ $\Theta(n^6)$

☐ $\Theta\left(\frac{2n}{n^6}\right)$

☐ $\Theta(n^2)$

☐ $\Theta(2n \lg 6)$

correct answer solve by hadi

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Question No : 17 of 26

Marks: 1 (Budgeted Time 1 Min)

```
1  do maximal ← true
2  for j ← 1 to n
3      do
4          if (i ≠ j) & (P[i].x ≤ P[j].x) & (P[i].y ≤ P[j].y)
5              then maximal ← false break
```

Answer (Please select your correct option)

- ☐ $\Theta(n^6)$
- ☐ $\Theta\left(n^{\frac{2n}{6}}\right)$
- ☐ $\Theta(n^2)$
- ☐ $\Theta(2n \lg 6)$

correct answer solve
by hadi

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Question No : 17 of 26

Marks: 1 (Budgeted Time 1 Min)

```
3   for j ← 1 to n n times
4   do
5       if (i ≠ j) & (P[i].x ≤ P[j].x) & (P[i].y ≤ P[j].y) 4 accesses
6           then maximal ← false break
7   if maximal
```

Answer (Please select your correct option)

☐ $\Theta(n^6)$ ☐ $\Theta\left(n^{\frac{2n}{6}}\right)$ ☐ $\Theta(n^2)$

same back,,,,correct answer solve by hadi

☐ $\Theta(2n \lg 6)$

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Question No : 17 of 26

Marks: 1 (Budgeted Time 1 Min)

```

5      if (i ≠ j) & (P[i].x ≤ P[j].x) & (P[i].y ≤ P[j].y) 4 accesses
6      then maximal ← false break
7      if maximal
8      then output P[i].x, P[i].y 2 accesses

```

Answer (Please select your correct option)

☐ $\Theta(n^6)$
☐ $\Theta\left(n^{\frac{2n}{6}}\right)$
☐ $\Theta(n^2)$

same back,,,,correct
answer solve by
hadi

☐ $\Theta(2n \lg 6)$

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Question No : 18 of 26

Marks: 1 (Budgeted Time 1 Min)

Which one is the correct conversion of the given algorithm to sigma notation?

MAXIMA(int n, Point P[1...n])

```
1 for i ← 1 to n  
2 do maximal ← true
```

Answer (Please select your correct option)

☐ $\sum_{i=1}^n 4n + 4$

☐ $\sum_{i=1}^n 4n + 2$

☐ $\sum_{i=1}^n 4n + 2n$

☐ $\sum_{i=1}^n 2n + 2$

correct answer solve
by hadi

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Question No : 18 of 26

Marks: 1 (Budgeted Time 1 Min)

```
1 for i ← 1 to n n times
2 do maximal ← true
3   for j ← 1 to n n times
4   do
5     if (i ≠ j) & (P[i].x ≤ P[j].x) & (P[i].y ≤ P[j].y) 4 accesses
```

Answer (Please select your correct option)

☐ $\sum_{i=1}^n 4n + 4$

☐ $\sum_{i=1}^n 4n + 2$

☐ $\sum_{i=1}^n 4n + 2n$

☐ $\sum_{i=1}^n 2n + 2$

correct answer solve
by hadi

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Question No : 18 of 26

Marks: 1 (Budgeted Time 1 Min)

```
3   for j ← 1 to n   n times
4   do
5       if (i ≠ j) & (P[i].x ≤ P[j].x) & (P[i].y ≤ P[j].y)   4 accesses
6           then maximal ← false break
7   if maximal
```

Answer (Please select your correct option)

☐ $\sum_{i=1}^n 4n + 4$

☐ $\sum_{i=1}^n 4n + 2$

☐ $\sum_{i=1}^n 4n + 2n$

☐ $\sum_{i=1}^n 2n + 2$

correct answer solve
by hadi

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Question No : 18 of 26

Marks: 1 (Budgeted Time 1 Min)

```
5      if (i ≠ j) & (P[i].x ≤ P[j].x) & (P[i].y ≤ P[j].y) 4 accesses
6      then maximal ← false break
7  if maximal
8  then output P[i].x, P[i].y 2 accesses
```

Answer (Please select your correct option)

☐ $\sum_{i=1}^n 4n + 4$

☐ $\sum_{i=1}^n 4n + 2$

correct answer solve
by hadi

☐ $\sum_{i=1}^n 4n + 2n$

☐ $\sum_{i=1}^n 2n + 2$

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Question No : 19 of 26

Marks: 1 (Budgeted Time 1 Min)

When this relation will be true? $f(n) = O(f(n)^2)$

Answer (Please select your correct option)

- ☒ Always True
- ☐ Never True
- ☐ Some times True only
- ☐ True for small values of n



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Question No : 20 of 26

Marks: 1 (Budgeted Time 1 Min) 

Which of the following functions grows fastest as n grows larger?

Answer (Please select your correct option)

- ☐ n^2 correct answer solve by hadi
- ☐ $\log n$
- ☐ $n \log n$
- ☐ 2^n

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